

Multi-Source Candidate Data Transformer — Technical Design

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1 · Pipeline / Step Breakdown

Stage	Responsibility	Output
1 Ingest	Detect source type by extension; skip missing/corrupt files without crashing.	raw payload
2 Extract	CSV: DictReader + header alias. JSON: key-mapping walker. PDF/DOCX: pdfplumber/python-docx + regex sections. TXT: rule-based patterns.	List[IntermediateRecord]
3 Normalize	Phone->E.164, date->YYYY-MM, country->ISO-3166-a2, email lowercase+validated, skills->alias map.	List[NormalizedRecord]
4 Match & Group	Email -> phone -> name+company (fallback only, never a parallel key). See section 3.	Dict[candidate_id -> records]
5 Merge	Priority order ATS>CSV>Resume>Notes. Scalars: highest-priority wins. Arrays: union+dedup. Structs: key-merge + gap-fill.	CanonicalRecord
6 Confidence	per_field = source_weight x agreement_bonus. See section 3 example.	confidence map
7 Project	Runtime config selects/renames/normalizes fields. Pure transform -- canonical record never mutated.	ProjectedRecord
8 Validate	Schema check before emit. Required-missing -> on_missing policy (null/omit/error).	Final JSON

2 · Canonical Output Schema & Normalization

Field	Type	Normalization Rule
candidate_id	string	SHA-256(lowercase primary email). Fallback: name+phone, then bucket key.
full_name	string null	Title-case, whitespace-stripped.
emails	string[]	Lowercase, RFC-5322 validated, deduped.
phones	string[]	E.164 (+CC+digits). Invalid -> null, never stored.
location	{city,region,country}	country -> ISO-3166 a2 via lookup table.
links	{linkedin,github,portfolio,other[]}	URL normalized to https://, deduped.
headline	string null	Max 160 chars, stripped.
years_experience	number null	Explicit if present; else derived only if ALL experience entries have dates.
skills	string[]	Canonical lowercase via alias map ('js'->'javascript').
experience	{{company,title,start,end,summary,location?,employment_type?}}	Dates YYYY-MM; newest-first; deduped by company+title.
education	{{institution,degree,field,end_year}}	end_year integer; deduped by institution+degree.
confidence	{field->float}	Per-field score 0-1, stored separately from provenance.
provenance	{field->{source,method}}	Field-wise object (O(1) lookup), not an array.
metadata	{processed_at,merged_from[],version}	ISO-8601 timestamp; source list; schema version.
overall_confidence	float	Weighted mean of all non-null field confidence scores.

3 · Matching & Merge Policy

Identity matching (strict precedence):

- normalized email -- checked first
- normalized phone -- used if no email
- name + company -- used ONLY if record has neither email nor phone (never a parallel key, to avoid false merges)

Source priority (descending):

ATS JSON 0.90 > CSV 0.75 > Resume 0.70 > Notes 0.50

Conflict resolution:

- Name: choose the most complete valid name.
- Other Scalars: highest-priority non-null wins; ties -> most recent, else first occurrence.
- Arrays: full union with deduplication.
- Structured fields(experience, education): key-merge, gap-filled from lower-priority sources.

Different fields require different merge strategies instead of one common priority rule.

Confidence example:

full_name in ATS(0.90)+CSV(0.75), agree:

conf = 0.90 x 1.0 = 0.90

Same field, only CSV, no agreement:

conf = 0.75 x 0.8 = 0.60

During evaluation, confidence explanations may be emitted. In production, only numeric scores would typically be stored.

Core principle: if a value can't be determined confidently, return **null** -- never guess. Wrong-but-confident pollutes hiring decisions.

5 · Edge Cases & Deliberate Descopes

Edge Case	Handling
Corrupt or unreadable input	try/except per extractor; source skipped + logged; pipeline continues.
Invalid phone format	Regex fails E.164 check -> null, provenance method='parse_failed'.
Unsupported date format	8-pattern library covers common cases; unrecognized -> null, never guessed.
Two people share name + company	Weak fallback key fires ONLY when a record has no email and no phone -- never registered alongside a strong key, so distinct people with unique emails/phones can never be falsely merged just by sharing a name+employer.
Required field missing	on_missing policy applied; hard error raised only if on_missing='error'.

Deliberate descopes (time): GitHub API source dropped in favor of deeper resume/notes coverage · fuzzy name dedup (Levenshtein) not implemented -- exact normalized match only · English-language sources only · no persistent database -- JSON file output per run.

4 · Runtime Config & Projection

Projection is a **pure read-only transform** of the canonical record -- the config never modifies the canonical schema itself. This separation allows the same canonical profile to be projected into multiple output schemas without recomputing the merge pipeline.

Config capabilities:

- fields[]** -- select output fields; dot-path + array index (emails[0]).
- from** -- rename: canonical path -> output key.
- normalize** -- re-apply E164/canonical at projection time.
- include_confidence / include_provenance** -- strip blocks if false.
- on_missing** -- null (default) / omit / error.

Projection flow:

```
canonical -> project(config) -> projected_dict ->
validate(schema) -> emit
```

The canonical record is generated exactly once and remains unchanged throughout the pipeline; only the projection layer adapts the output to different consumer requirements.

Validation: output checked against a schema built from the config (Pydantic-style/inline JSON Schema). required:true failures trigger the on_missing policy; type mismatches raise before return.